Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform 4-space indentation used throughout all Python files	Yes	PEP8 standard followed
2	Proper File Structure	Files organized into: app.py, templates/, model/, dataset/, notebooks/, plots/	Yes	Template structure followed
3	Meaningful Variable Names	Variables: Development, X, y, x_train, x_test, y_pred, reg clearly reflect purpose	Yes	Descriptive naming used
4	Function / Method Names	Flask routes: home(), prediction(), my_home(), predict() are descriptively named	Yes	RESTful naming convention
5	Code Comments	Inline comments like #Importing the dataset, #Train and test split used throughout	Yes	Comments on every major step
6	Modular Design	ML pipeline separated into epics (epic2–epic7), Flask app in app.py, templates separate	Yes	Epic-based modular structure
7	No Redundant Code	No duplicate code; preprocessing reused across epics via shared notebook sections	Yes	DRY principle applied
8	Error Handling	Flask predict() route handles out-of-range HDI predictions with else clause	Partial	Can add try-except blocks

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	HDI.pkl (trained model)	Python / Scikit-learn Pickle	Loaded in app.py for every /predict POST request	High
2	LabelEncoder (le)	Scikit-learn	Applied in Epic 4 preprocessing + expected in Flask input	High
3	X.fillna(X.mean())	Pandas	Applied consistently after each feature selection step	High
4	predict() Flask route	Python / Flask	Core route handling all prediction requests via POST	High
5	Jinja2 {{prediction_text}}	Jinja2 Template	Used in resultnew.html to display dynamic prediction result	High

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5/5	Clean Epic-based modular structure with clear file organization
Readability	4/5	Meaningful names and comments throughout; can add more docstrings
Reusability	5/5	Pickle model reused across prediction requests; features reused in pipeline
Documentation / Comments	4/5	Inline comments on every major code block; README provided
Overall Score	4.5/5	High-quality, maintainable, production-ready codebase